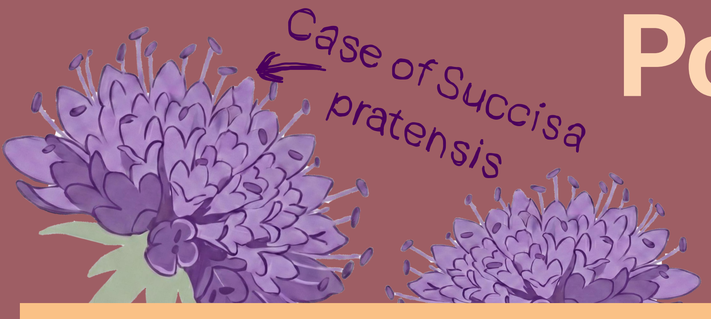
Pollen transfer efficiency

What decides upon deposition?

Petr Švanda, Jakub Štenc, Martin Freudenfeld, Eva Matoušková





INTRODUCTION

Understanding the link between **vegetation composition**, **plant-pollinator interactions**, and actual **pollen deposition** is crucial for assessing reproductive success in plants. While vegetation sets the stage, the foraging behavior and constancy of specific pollinators often dictate the efficiency of pollen transfer.

METHODS

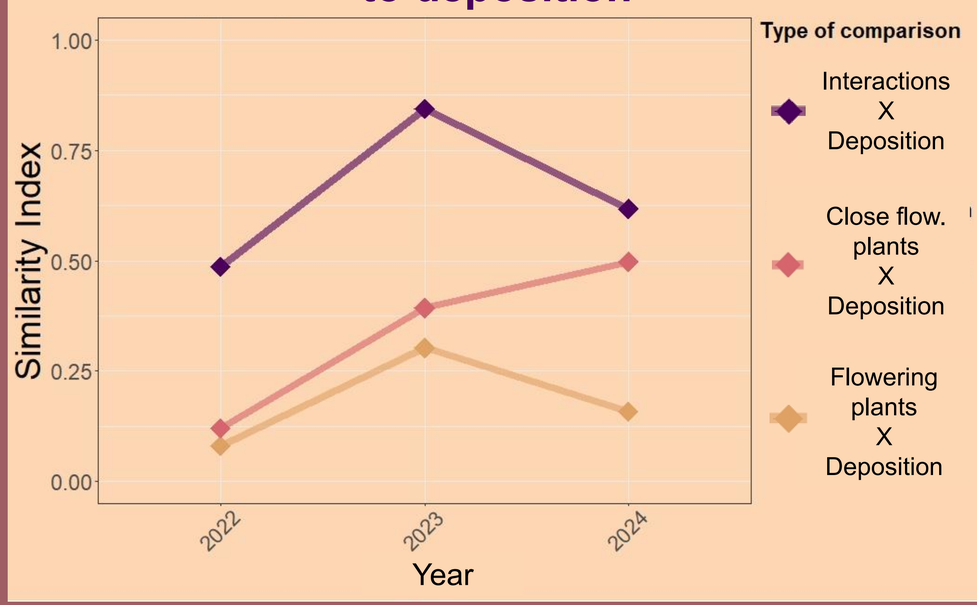
- 93 permanent plots for flowering plant survey
- Sampling of plant- pollinator interactions around the same plots for interaction data
- Sampling of mature stigmata from *Succisa pratensis* for total depositions
- Sampling single visit deposition from *Succisa pratensis*

INFO

Pollination ecology group website →
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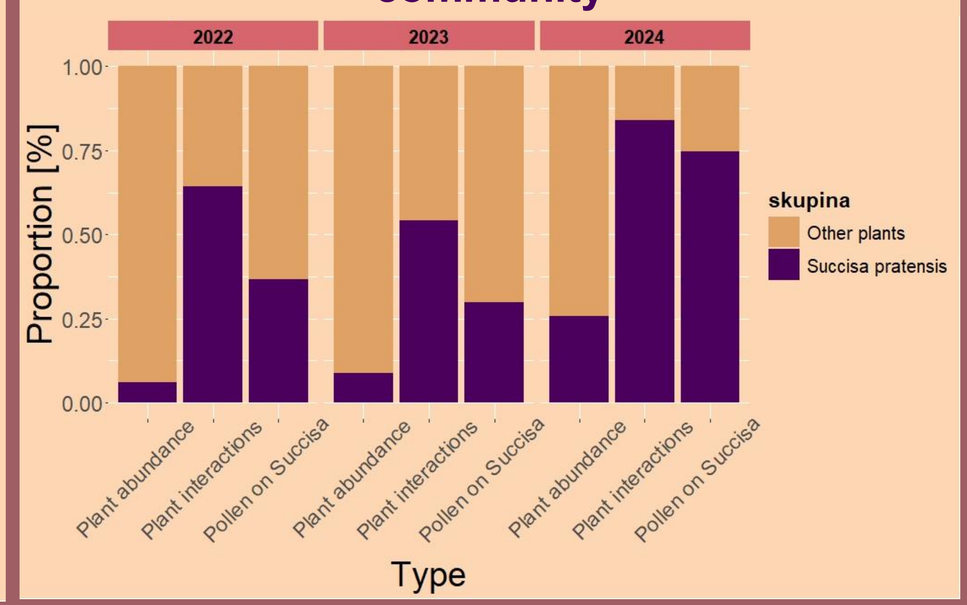
Charles university, Department of botany

Similarity of flowering plants and interactions to deposition



- Plant-pollinator interactions are better fit as a predictor for species composition of deposition
- Suprisingly the close flowers predicts pollen composition better which could be explained by pollinator fidelity/foraging pattern/...
- The 2024 spike is caused by increase of *Succisa* abundance, but more about that later

Proportion of Succisa pratensis in the community



- The Increase in *Succisa* abundance translates into both a rise in conspecific interactions* and deposition
- The right proportion of heterospecific/conspecific deposition is suited somewhere between the abundance and interactions as the interactions increase disproportionaly with abundance

Similarity of flowering plants

0.69	0.7	0
0.65	0	0.7
0	0.65	0.69

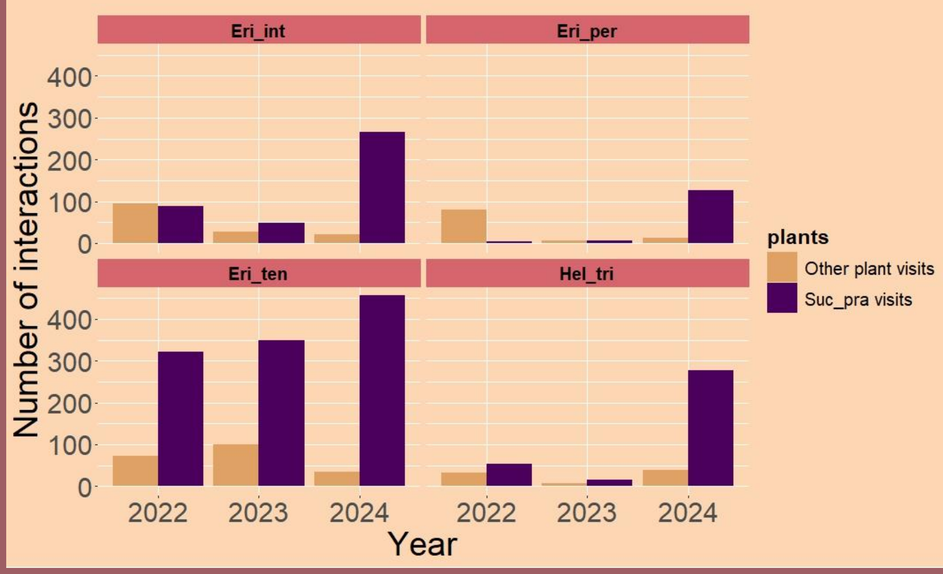
Similarity of interactions

0.96	0.86	0
0.94	0	0.86
0	0.94	0.96

- Yearly changes in the plant composition (and pollinators* 0.96 - 0.77) does not translate into interactions (only using plots where *Succisa* was present)
- Indicating stable/fidelty group of *Succisa* pollinators

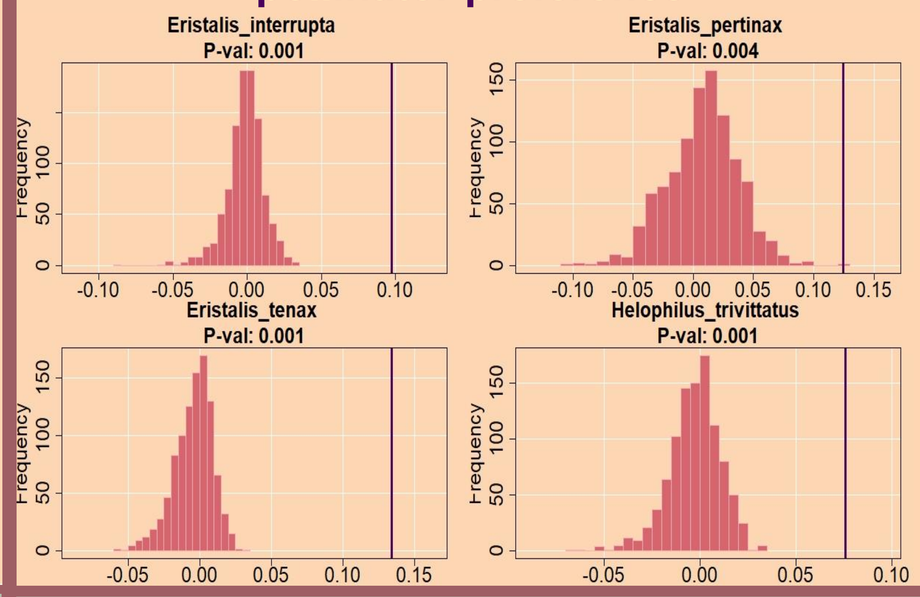


Proportion of visits in dominant Succisa pratensis pollinator species



- The most frequent and stable pollinator for *Succisa* is *Eristalis tenax* which always prefer *Succisa* over other plants
- *Eristalis pertinax* and *Eristalis interrupta* decide upon abundances of a given species
- This could also be said about *Helophilus trivittatus*, but his own abundance spiked at 2024 which could be the cause of change

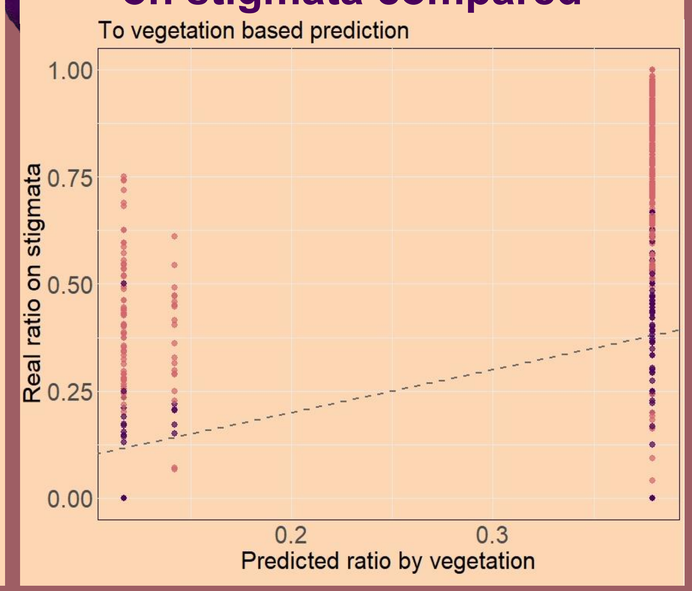
Effect of Succisa abundance increase on pollinator preference



Randomization model comparing real and generated abundance ~ visits slopes

- All dominant pollinator species of *Succisa* are driven by its flowering abundance, with tenax being the most constant
- Abundances for heterospecific (other) plants affected visits the same way, but with weaker effect (0.03-0.07), indicating preference towards *Succisa*

Real proportion of C/H pollen on stigmata compared



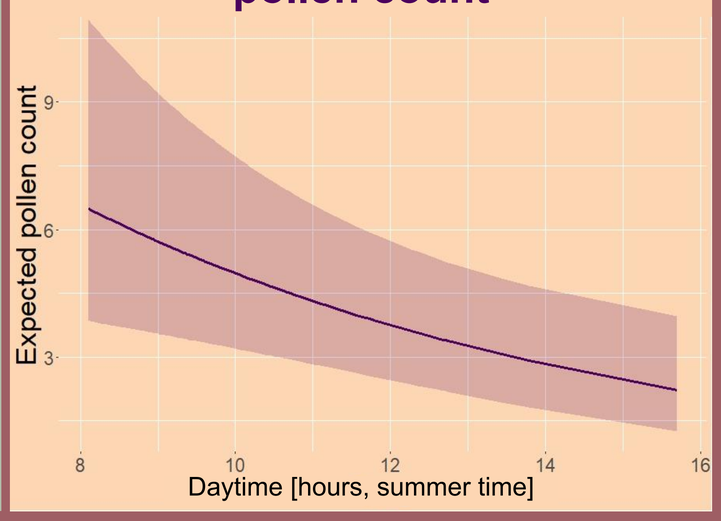
Model comparing real C/H deposition to generated stigmata using flowering plant abundances, also done for interactions. Finally comparing error rate of each prediction

- Suprisingly at con/heterospecific level flowering plants based prediction showed lower error rate than the interaction one

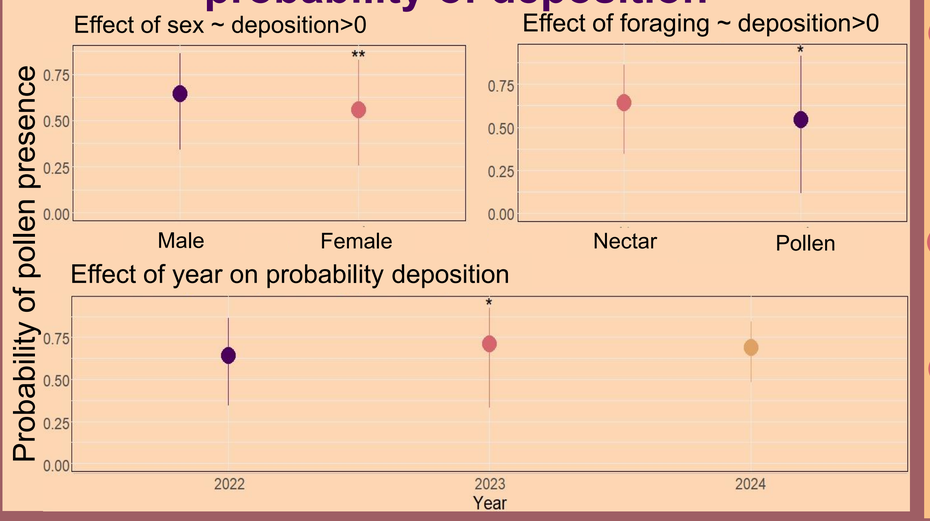
SINGLE DEPOSITION CORNER

Results of hurdle zero-inflated model using truanced poisson distribution for count part and negative binomial for the zero part. Daytime was only significant factor in count part. Sex, foraging and year were significant in the zero-part.

Effect of daytime on deposition pollen count



Influence of sex, foraging and year on probability of deposition



CONCLUSION

- Species composition of deposited pollen has higher similarity with plant-pollinator interactions than vegetation.
- Interactions in relevés (with *Succisa*) remain stable even with changes in vegetation and pollinators. Which is mainly driven by constancy and stability of *Eristalis tenax*.
- However *E. tenax* is also driven by flower abundance as well as the other dominant *Succisa* pollinators.
- In single deposition, pollinators sex and foraging of a visit decides upon succes of a deposition and time about how succesful the deposition will be.